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I am an MS-trained data scientist with expertise in data analysis, AI/machine learning, and data visualization, and I bring a strong programming foundation in R, Python, and SAS. While my academic background and professional experience are rooted in biomedical data science, I am seeking opportunities to apply my skills more broadly across diverse industries.

At Cincinnati Children's Hospital, I worked as a Bioinformatics Analyst on NIH-funded spatial transcriptomics projects, applying statistical and computational methods to study gene expression in lung tissue. During my Master's research, I developed and applied machine learning models to investigate gene regulatory networks linked to Alzheimer's disease, contributing to a manuscript under journal revision. These projects strengthened my ability to translate complex data into actionable insights.

In addition to my biomedical work, I have cultivated strong technical skills across programming and visualization platforms. I am proficient in Python (Pandas, NumPy, Matplotlib, Scikit-learn, SciPy) and R (Tidyverse, ggplot2, dplyr, igraph, visNetwork, Shiny, Seurat), with additional experience in SAS, SQL, Java, and MATLAB. I regularly use these tools to design statistical workflows, build predictive models, and create clear and interactive data visualizations.

I am eager to contribute these versatile skills to an organization where data-driven insights drive innovation, whether in healthcare, technology, business, or beyond. I would welcome the opportunity to discuss how my background in analysis, machine learning, and programming can add value to your team.

Thank you for considering my application.

Best regards,
Ryan C. Burczak

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RYAN CONWAY BURCZAK

EDUCATION:

May 2024 **University of Wisconsin**, School of Medicine and Public Health, Madison, WI
M.S., Biomedical Data Science

May 2021 **Colgate University**, Hamilton, NY
B.A., Computer Science

SUMMARY OF QUALIFICATIONS:

- Experienced in programming languages: Python, R, Java, JavaScript, HTML, SQL, MATLAB
- Python experience for data science: Pandas, NumPy, Matplotlib, Scikit-learn, SciPy
- R experience: Tidyverse, ggplot2, dplyr, igraph, visNetwork, Shiny, Seurat, plotly
- Proficient in data cleaning and transformation, data integration, and data visualization
- Experienced in statistics, machine learning methodologies, and image analysis/processing

WORK EXPERIENCE

2025 - Present Data Science Consultant Feinberg School of Medicine Northwestern University

- Overhauling bulk RNA-seq and scRNA-seq data visualization apps for the Winter Lab of Macrophage Genomics to significantly improve speed, reliability, and analytical capabilities

2024-2025 Bioinformatics Analyst and Web Developer Division of Pulmonary Biology
Cincinnati Children's Hospital Medical Center, Cincinnati Ohio

- Contributed to the development of the LAM Cell Atlas by creating graphs for gene regulatory networks and cell interactions in lymphangioleiomyomatosis (LAM) lung tissue, using R Shiny.
- Learned about and applied the single cell analytical pipeline for spatial transcriptomics to study the gene networks controlling idiopathic subglottic stenosis (iSGS) disease.
  - Used the Seurat analytical package for scRNA-seq spatial transcriptomics (QC, cleaning, normalization, scaling and integration of datasets. Followed by clustering, UMAP for dimensionality reduction and visualization, and identifying differentially expressed genes in cell type clusters.)
  - Used CCA, RPCA and Harmony methods of integration for datasets.
- Provided data visualization support for the lab group using various techniques such as dimensionality reduction (PCA, UMAP and tSNE), heatmaps, network graphs, Venn diagrams and other plots.
- Developed and maintained the LAM Cell Atlas website and LGEA Web Portal using Javascript

2023-2024 M.S. Graduate Research Data Science Core Laboratory University of Wisconsin

- Applied machine learning methods to analyze multimodal data for generating genotype-phenotype predictive models and to understand the genomics of Alzheimer's Disease and schizophrenia.
- Gained research experience with data cleaning and preprocessing, hyperparameter tuning for multiple machine algorithms (such as k-nearest neighbors, logistic regression, support vector machine, stochastic gradient descent, extreme gradient boosting, neural networks, and random forest), various metrics for model performance, and data visualization.
- Optimized and compared nine different machine learning methods to develop predictive models for Alzheimer's Disease, using cross validation and eight different objective metrics
- Created data visuals for computed data using ggplot2 and Pandas packages
- Used Integrated Gradient and SHAP algorithms to rank high-dimensional features for importance/ classification influence; improved all nine machine learning models
- Used bash scripting and compared PLINK, PRSice and LDPred-2 algorithms to determine polygenic risk score for schizophrenia
- Created an interactive web application for the Wang lab to visualize data using R Shiny, igraph, and visNetwork packages

WORK EXPERIENCE:

2022 Graduate Summer Research Center for Research Informatics University of Chicago

- Analyzed and visualized cell-to-cell communication in complex cell systems via differentially regulated pairs of secreted proteins and their target receptors using Cytoscape software platform
- Developed an interactive client-server application using R Shiny to visualize and analyze genomic data from next-generation sequencing and gene expression analysis

2019 Undergraduate Summer Research Institute for Quantitative Health Sciences/Engineering Michigan State University

- Performed image processing research in the Medical Imaging and Data Integration Lab
- Developed Python-based algorithms for image normalization and histogram equalization to prepare medical images for machine learning applications
- Used Pydicom to import X-ray images, NumPy to process the images, and Matplotlib to display the processed images

2018 Undergraduate Summer Bioinformatics Department of Medical Physics Royal College of Surgeons in Ireland

- Performed beta testing, debugging and modification of the user protocols for Cell DIVE tissue imaging and analysis platform at GE Global Research; defined edits and updates of user manual and training materials
- Transferred beta software from GE Global Research to collaborators at Royal College of Surgeons in Ireland; performed image analysis on joint US-Ireland research project to predict the effectiveness of anti-cancer therapies; developed an algorithm to identify cancerous cell types
- Commercial product launch of Cell DIVE by GE Life Sciences in Summer 2019

2015-2017 Student Research Assistant Gen*NY*Sis Center for Excellence in Cancer Genomics University at Albany

- Performed molecular biology research on breast cancer
- Utilized genomic computational tools and databases including the UCSC Genome Browser, NCBI GenBank search tool, EMBL SMART protein analysis tool, and NCBI Primer-BLAST search tool
- Developed and validated the first laboratory test for detecting BTK-52 messenger RNA in cancer cell lines
- Received the Superior Presentation Award at the 2016 Sigma Xi Student Research Conference, Atlanta

COLLEGIATE INVOLVEMENT:

Clubs

- Colgate Coders
- Students for Environmental Action
- Newman Community
- Ski and Snowboard Club

Music

- University of Wisconsin Medical Sciences Orchestra
- Colgate Chamber Players
- Colgate Symphonic Band
- Instrument proficiency with the piano, French horn, and harpsichord